

AXIOM

Zero-Trust Invariant & Blast-Radius Firewall for AI Dev Partners

Autonomous AI agents generate multi-file code in seconds, but introduce silent, unstated assumptions that bypass human review. Axiom acts as an adversarial auditor that uncovers ghost invariants, breaks them with synthesized stress tests, and maps downstream blast radius before merging to production.

THE PROBLEM

Confirmation Bias & Blindspots

AI code passes basic happy-path unit tests while hiding critical domain and boundary invariants.

THE MECHANISM

Adversarial Counter-Agent

Synthesizes property-based fuzz tests to actively smash assumptions and auto-guides IBM Bob to self-heal.

ENTERPRISE VALUE

in-toto DSSE Code Passport

Cryptographically attests code safety and blast radius containment for SOC2 and ISO compliance.

The Unwritten Assumption Blindspot

BOUNDARY NEGLECT

Negative Input Exploit

AI assumes transfer amount is strictly positive. A negative value causes balance to increment artificially (infinite credit duplication).

TENANT ISOLATION

Cross- Tenant Leakage

AI omits organization workspace boundaries. Rogue accounts can manipulate wallets and data across unrelated corporate tenants.

CONCURRENCY RACES

Non-Atomic Mutations

Read-modify-write patterns without serialized locks collapse under concurrent traffic, enabling double-spend vulnerabilities.

IDEMPOTENCY GAP

Duplicate Side-Effects

Missing replay token filtering causes network retry packets to debit accounts repeatedly for a single real transaction.

The Human Reviewer Paralysis:

When an AI dev partner generates a 500-line multi-file pull request, human engineers lack the cognitive bandwidth to trace unwritten edge-case assumptions. They rubber-stamp the PR because the code compiles cleanly—allowing catastrophic vulnerabilities into enterprise production.

Axiom: Zero-Trust Verification Engine



ADVERSARIAL SYNTHESIS

Property-Based Fuzzing Sandbox

Rather than generating compliant tests, Axiom synthesizes hostile edge-case payloads (negative floats, null bytes, concurrency spikes) targeting the exact extracted invariants.

AUTOMATED REMEDIATION

Closed-Loop Self-Healing with Bob

When invariants break, Axiom feeds the failure trace back to IBM Bob to synthesize an atomic zero-trust patch, re-verifies all tests, and cryptographically signs the PR.

FinTech Wallet Transfer Case Study

PHASE 1: INITIAL AI CODE

Happy-Path Illusion

IBMBob writes
`transfer_funds()`.
Standard unit tests pass
(100% green).

- 4 Ghost Invariants Live
- PostgreSQL Ledger at Risk
- Risk Score: 88/100 (Critical)

PHASE 2: ADVERSARIAL ATTACK

Vulnerabilities Exposed

Axiom executes
synthesized edge tests
live in the integrated
sandbox.

- Result: 3 FAILED / 0 PASSED
- Amount: -500.0 USD Exploit
- Balance incremented to 600!

PHASE 3: BOB AUTO-PATCH

Zero-Trust Verified

Failure stack traces
routed to Bob; zero-trust
invariant patch applied.

- Result: 3 PASSED / 0 FAILED
- Topology turns Emerald Green
- Risk Score: 12/100 (Secured)

Enterprise Impact:

Axiom issues an **in-toto / DSSE cryptographic Code Resilience Passport**. The PR description receives a verifiable markdown audit badge, giving enterprise engineering directors concrete proof of resilience before merging.

Technical Execution & Why We Win

FULL-STACK ARCHITECTURE

Built for Speed & Reliability

- **Backend:** Python 3.14, FastAPI, AST Analysis, pytest subprocess runner
- **Frontend:** React 18, Vite, Tailwind CSS, animated topology node graphs
- **AI Partner:** IBM Bob 2.0 / watsonx for code synthesis and self-healing
- **Compliance:** in-toto DSSE SHA-256 cryptographic attestation
- **Deliverable:** 100% open-source on GitHub with 1-click launcher

JUDGES SCORING BLUEPRINT

Why Axiom Takes the Podium

- **Uncrowded Space:** 90% build chatbots; Axiom solves automated enterprise verification.
- **High Visual Drama:** Red-to-green topology shifts and live pytest execution on camera.
- **IBM Ecosystem Fit:** Directly complements IBM Bob, turning it into a trusted enterprise tool.
- **Measurable Results:** Proven risk delta from 88/100 (critical) down to 12/100 (safe).

Ready for Production Evaluation

Public Repository:

<https://github.com/suwalkaanmol/axiom>

SOLO DEVELOPER: ANMOL
SUWALKA

